

Nikhil Dattu Bhorkade

Robotics Software Engineer

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ABOUT ME

Robotics & Automation Engineering graduate with hands-on experience in ROS 2, autonomous navigation, SLAM, Nav2, and robotic simulation. Passionate about developing intelligent robotic systems by integrating perception, localization, motion planning, and control for real-world automation applications.

EDUCATION

B.E. in Robotics and Automation Engineering, CGPA: 8.10

2026

Zeal College of Engineering and Research, Pune

TECHNICAL SKILLS

- **Robotics & Simulation:** ROS 2, SLAM, Autonomous Navigation, Gazebo, RViz 2, Nav2, TF2, MoveIt, URDF, RoboAnalyzer, Cartographer
- **Programming:** Python, C++
- **Design & CAD:** Fusion 360, SolidWorks, MATLAB
- **Hardware & Integration:** Embedded Systems, Control Systems, Sensor Integration, Hardware Prototyping
- **Operating Systems:** Ubuntu Linux, Windows, Raspbian Linux
- **Soft Skills:** Self-Learning, Quick Adaptability

PROFESSIONAL EXPERIENCE

Robotics Engineer Intern

Jan 2026 – Jun 2026

Kody Technolab Limited, Ahmedabad

- Developed and deployed ROS 2-based navigation systems for real and simulated mobile robots using Ignition Gazebo Fortress, validating performance in both virtual environments and physical hardware.
- Developed ROS 2 Nav2-based autonomous navigation by performing parameter tuning, optimizing multiple planners and controllers, and refining localization. Built 2D maps using SLAM and Cartographer, and implemented AMCL, SLAM-based, and Cartographer localization for reliable robot navigation.

ROS Developer Intern

Sep 2025 – Oct 2025

NISHCorp Technologies LLP, Mumbai

- Contributed to research and development activities for ROS 2-based robotic platforms.
- Performed robot simulation and controller testing using Gazebo, RViz, and ros2_control.
- Created and maintained technical documentation for simulation and validation.

PROJECTS

DuoSanBot: An Intelligent Dual-Mode Sanitization Robot with Autonomous Navigation

- Developed an autonomous sanitization robot using ROS 2, Nav2, and Gazebo, enabling indoor autonomous navigation and obstacle avoidance.
- Designed the robot using URDF and visualized navigation, sensor data, and robot states in RViz 2.
- Integrated ultrasonic sensors, ROS 2 nodes, and a dual sanitization system with a disinfectant pump and UV-C module for autonomous operation.
- Built a modular architecture supporting future integration of LiDAR, SLAM, AMCL, and IoT-based remote monitoring.

Factory Automation Simulation Dashboard

- Developed a ROS 2-based factory automation simulation with robot spawning, autonomous navigation, and industrial environment modeling.
- Built a Python-based GUI dashboard to visualize robot status, sensor health, system health, and emergency stop controls.
- Integrated Nav2 for autonomous navigation and ros2_control for simulated robot controller management.
- Monitored real-time robot telemetry and health information through ROS 2 topics and services.
- Tested and validated navigation and safety features in Gazebo and RViz.

ACHIEVEMENTS

- Qualified for the Aavishkar Zonal Level Project Competition (2025) for the development of DuoSanBot, an intelligent dual-mode sanitization robot featuring autonomous navigation.

CERTIFICATIONS & TRAINING

- ROS 2 Framework, Robotics (2025)
- ROS 2: Hardware Integration and ros2_control, Udemy (2025)
- Python Programming, Deutsche Bank CSR & Rubicon Foundation
- Pneumatic Controls & SMC Training Center, Pune